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Theorems of Boolean Algebra

Identity

$A + 0 = A$

$A \cdot 1 = A$

Complement

$A + \overline{A} = 1$

$A \cdot \overline{A} = 0$

Commutative

$A + B = B + A$

$A \cdot B = B \cdot A$

Associative

$A + (B + C) = (A + B) + C$

$A \cdot (B \cdot C) = (A \cdot B) \cdot C$

Distributive

$A \cdot (B + C) = A \cdot B + A \cdot C$

$A + (B \cdot C) = (A + B) \cdot (A + C)$

Null Element

$A + 1 = 1$

$A \cdot 0 = 0$

Involution

$\overline{(\overline{A})} = A$

Indempotence

$A + A = A$

$A \,.\, A = A$

Absorption

$A + (A \cdot B) = A$

$A \cdot (A + B) = A$

3rd Distributive

$A + \overline{A} \cdot B = A + B$

De Morgan's

$\overline{A + B} = \overline{A} \cdot \overline{B}$

$\overline{(A \cdot B)} = \overline{A} + \overline{B}$